Mini Test, III Sem, 2023-24, EE

Total Marks : 10 Time : 50 minutes

Let a be the last two digits of your scholar number

Q1. Given the function $e^{-x} + a\cos(3x)$, find a positive root of this equation correct to three decimal places. You may use any method or a combination of different methods.

Q2. Find the point (p, q) with $p > 0$ where the curves $y + a\sin\left(\frac{\pi}{6}\right)x^2 = e^{\frac{3}{2}}$ and $y = \tan\left(\frac{\pi}{3a}\right)x - \sqrt{\pi}$ intersect using the Newton Raphson method. The answer should be correct to three decimal places.

Q3. The displacement of a body in metres with respect to time in seconds is given as $s(t) = \frac{4}{30}x^6 - x^5 + x^4 - x^3 - \frac{19}{2}x^2$. Find the time $(t > 0)$ at which the acceleration of the body is a m/s^2, correct to three decimal places.